

Author	QC Manager Approval	Head of Supply Chain Approval
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ATR No. None

1. SOURCE DOCUMENT(S) AND DATE(S):

Documents Names	Documents Numbers	Effective Dates
pH method	ATM-BK-001	
Moisture Analyzer method	ATM-BK-002	
Assay HPLC method	ATM-BK-003	

2. EHS:

- 1.1 As general recommendations, avoid all contact with reagents.
 1.2 Always use suitable personal protective equipment where appropriate.
 1.3 Nicotine is fatal if inhaled, has contact with skin, swallowed, and is toxic to aquatic life with long-lasting effects.

Nicotine



Organic solvents



Tetrahydrofuran



3. Product Information:

#	Product	Batch#	Type of sample	Received Date	COA Number
1	Icy Rush - Iced Mint -10 mg	12121	None	2026-05-18	None



4. Test to be performed at Release:

Test name	Acceptance criteria
Appearance	N/A
Assay	60.0 - 120.0
Average Fill weight	295.0 - 355.0
Average gross weight per pouch	340.0 - 400.0
Description	N/A
Identification	N/A
Length	27.0 - 32.0
Moisture	≤ 0.2
Odour	N/A
Water Content LOD	≥ 30.0
Width	12.0 - 14.0
pH	7.5 - 10.5

5. Testing specific requirements:

Equipment's:	Instrument Code	Calibration Due date
HPLC		
pH Meter		
Analytical Balance		

Chemicals:	
Ammonium hydroxide	Ammonium acetate
Acetonitrile	Tetrahydrofuran
Methanol	Water (De-ionized or HPLC quality)
	Nicotine Standard



6. Testing Procedure:

6.1. Appearance and Odour

Appearance	Appearance satisfactory and complies with Specs
Odour	Odour satisfactory and complies with Specs

#	Appearance	Odour	Done By/Date
12121	White, fine crystals	Almost odourless	Waleed



6.2. pH

Weigh out of 2.5g	Dissolve in a centrifuge tube 50 mL with purified water
Add a magnet and place the beaker on the magnetic stirrer.	
Place the pH electrode gauge in the beaker. Leave until the result has stabilized.	

Batch #	pH	Specification	Done By/Date
12121	8.0	7.5 - 10.5	Waleed

6.3. Moisture Analyzer

Moisture Analyzer Setting	
Temperature	105 °C
Time	10 min or until result stabilized.
Unit of measurement	M%
Weigh about 1.5 to 2 g of sample into the sample pan.	
Wait until the time is completed or the result is stable in the screen.	

Batch #	Moisture	Specification	Done By/Date
12121	0.002	≤ 0.2	Waleed



6.4 Average of Length and Width for wet pouches

Batch #	Length	Width	Done By/Date
12121	30.0	13.0	Waleed



6.5 Average gross weight/ Fill weight per pouch (mg) dry pouches

Batch #	Gross Weight	Fill Weight	Done By/Date
12121	350.0	300.0	Waleed



6.6. Assay test

6.6.1. Preparation of Mobile Phase A:

QC Unique Reference number DZRT- - -	Date of Preparation: 	Prepared By:
	Date of Expiry: 	
This Preparation is valid for 14 Days		
Volume Prepared: ()		
Into a Volumetric Flask, every 1000 mL accurately weighs 0,78 g ammonium acetate.		Add 750 ml methanol and mix until the ammonium acetate is dissolved and complete the volume
☐ Step Completed		
Adjust to 10 pH (± 0.1) With ammonium hydroxide. pH Value: _____		Filter this solution through a vacuum pump with a 47mm Filter with 0.45 µm pore.
☐ Step Completed		

6.6.2. Preparation of Mobile Phase B:

QC Unique Reference number DZRT- - -	Date of Preparation: 	Prepared By:
	Date of Expiry: 	
This Preparation is valid for 14 Days		
Volume Prepared: ()		
Into an Appropriate Flask, every 1000 mL accurately weigh 0,78 g ammonium acetate.		Add 750 ml Water and mix until the ammonium acetate is dissolved and complete the volume
☐ Step Completed		
Adjust to 10 pH (± 0.1) With ammonium hydroxide. pH Value: _____		Filter this solution through a vacuum pump with a 47mm Filter with 0.45 µm pore.
☐ Step Completed		



6.6.3. Preparation of Mobile Phase C:

Tetrahydrofuran (for “peak polish”) degas 15 min

QC Unique Reference number DZRT- - -	Date of Preparation: 	Prepared By:
	Date of Expiry: 	
This Preparation is valid for 28 Days		

6.6.4. Gradient Time Table:

	Time (min)	mobile phase A (%)	mobile phase B (%)	mobile phase C (%)
1	Initial	85,0	14,0	1,0
2	0.5	85,0	14,0	1,0
3	1.5	99,0	0	1,0
4	8.0	99,0	0	1,0
5	9.0	85,0	14,0	1,0
6	10	85,0	14,0	1,0

6.6.5. Chromatographic conditions:

Flow rate	1.0 mL/minute
Injection volume	10 µl
Run Time	Approximately 10 minutes
System Suitability Check:	
Standard Precision (n=5)	NMT 2.0%
STD Check	NMT 2.0%



6.6.6. Preparation of Solvent solution:

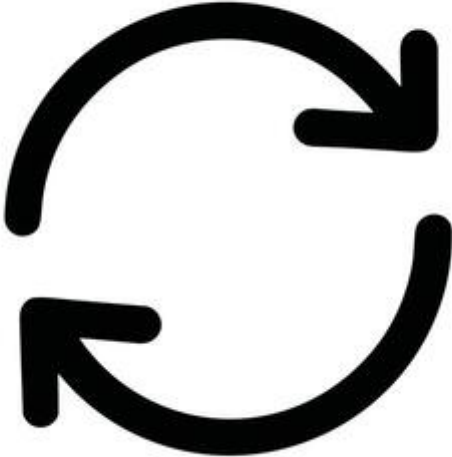
QC Unique Reference number DZRT- - -	Date of Preparation: 	Prepared By:
	Date of Expiry: 	
This Preparation is valid for 28 Days		
Volume Prepared:		
In a Suitable Container, mix (200 ml water and 800ml methanol)		

6.6.7. Nicotine Standard Solutions:

Nicotine Standard A QC Unique Reference number DZRT- - -	Date of Preparation: 	Prepared By:
Nicotine Standard B QC Unique Reference number DZRT- - -		
Date of Expiry: 		
This Preparation is valid for 14 Days at ambient conditions		

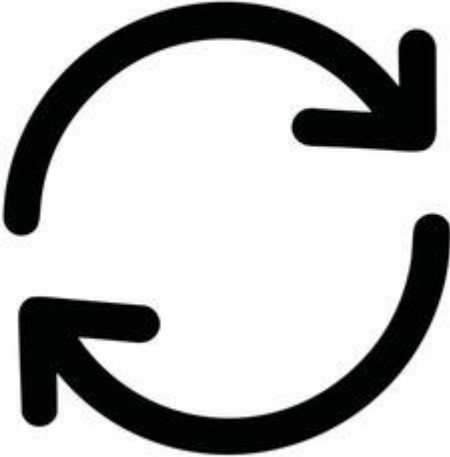
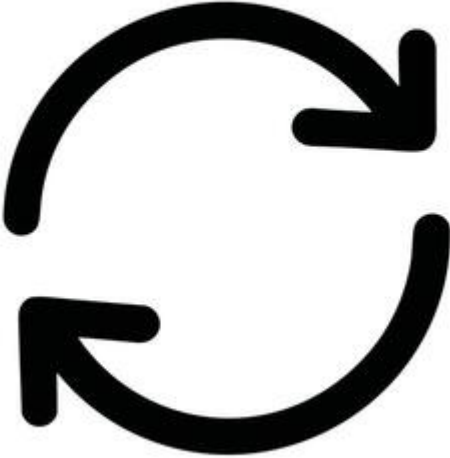
Standard Preparation					
Into a 100 mL Volumetric Flask, accurately transfer 1ml of the Nicotine Standard	Complete the volume by Solvent solution (Stock)		Into a 50 mL Volumetric Flask, accurately transfer 5ml from the Stock	Complete the volume by Solvent solution (Working)	



		Shake heavily for 30 seconds			Shake
					
using a syringe and filter to an HPLC vial.					



6.6.8. Preparation of Assay Sample Solutions:

☐ Blend	Into a 50ml centrifugal tube, Transfer the weight of 1000mg of the blend	add 50 ml of Solvent solution	Shake using a Vortex 	Let settle and separate
☐ Pouch	Into a 200ml plastic centrifugal tube, cut and transfer 20 pouches	add 100 ml of Solvent solution	Sonicate for 15 min, Stir for 5 min 	Complete the volume by Solvent
using a syringe and filter to an HPLC vial.				



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Sign/Date **Sign/Date**

